

1 Choice modeling

1.1 Binary Choice

We have

- $y = \text{observed}$ response variable, taking values 0 or 1 (No or Yes).
- $x_1, \dots, x_p$ attributes of choice, or, explanatory variables

We model the responses as

$$y^* = \beta_0 + \sum_{j=1}^p \beta_j x_j + \epsilon = \boldsymbol{\beta}^\top \mathbf{x} + \epsilon.$$

Here y^* is an unobserved latent variable, ϵ is the error term and we denote $\mathbf{x} = (1, x_1, \dots, x_p)$ (column vector) and $\boldsymbol{\beta} = (\beta_0, \dots, \beta_p)$ to be the (column) vector of $p+1$ parameters. The relationship between y and y^* is:

$$y = \begin{cases} 1 & \text{if } y^* \geq 0 \\ 0 & \text{otherwise} \end{cases}.$$

Now considering Y to be the random variable representing the response,

$$\begin{aligned} \Pr(Y = 1|\mathbf{x}) &= \Pr(Y^* \geq 0|\mathbf{x}) = \Pr(\boldsymbol{\beta}^\top \mathbf{x} + \epsilon \geq 0|\mathbf{x}) \\ &= \Pr(\epsilon \geq -\boldsymbol{\beta}^\top \mathbf{x}|\mathbf{x}) \\ &= \Pr(\epsilon \leq \boldsymbol{\beta}^\top \mathbf{x}|\mathbf{x}) \text{ (assuming } \epsilon \text{ is symmetric)} \\ &= F(\boldsymbol{\beta}^\top \mathbf{x}) \end{aligned}$$

where $\epsilon \sim F(\cdot)$.

With a choice of $F(z) = \frac{e^z}{1+e^z}, z \in \mathbb{R}$ we get,

$$\Pr(Y = 1|\mathbf{x}) = \frac{e^{\boldsymbol{\beta}^\top \mathbf{x}}}{1 + e^{\boldsymbol{\beta}^\top \mathbf{x}}}.$$

This is the logistic regression model we have seen in the previous class.

Also note that with a choice of $F(z) = \Phi(z), z \in \mathbb{R}$, where Φ denotes the standard normal CDF, we get,

$$\Pr(Y = 1|\mathbf{x}) = \Phi(\boldsymbol{\beta}^\top \mathbf{x}).$$

This is the **probit** regression model we have seen in the previous class.

1.2 Multinomial Choice

Given $k = 1, \dots, K$ choices (alternatives), $i = 1, \dots, n$ consumers (observations), the utility of consumer i for alternative k is,

$$U_{ik} = \boldsymbol{\beta}^\top \mathbf{x}_{ik} + \epsilon_{ik}.$$

Here $\boldsymbol{\beta}$ is the weight given to the observable information $\mathbf{x}_{ik}$ that includes aspects specific to the individual i and choice (alternative) k . The term ϵ_{ij} captures the noise term that models aspect of choice making not captured by attributes included in $\mathbf{x}_{ik}$.

For example x_{ik} might include demographic information, information such as price of a product. Then the probability that the i -th individual chooses the k -th alternative is given by

$$\Pr(Y_i = k) = \Pr(U_{ik} > U_{il} \forall l \neq k).$$

In some cases, you might have alternate specific constants ASC_k for each alternative.

$$U_{ik} = ASC_k + \boldsymbol{\beta}^\top \mathbf{x}_{ik} + \epsilon_{ik}.$$

Assuming that the ϵ_{ik} are independent and identically distributed following a Gumbel distribution,

$$\begin{aligned} \text{CDF : } F(y) &= e^{-e^{-y}}, \quad y \in \mathbb{R}, \\ \text{PDF : } f(y) &= e^{-y} e^{-e^{-y}}, \quad y \in \mathbb{R}. \end{aligned}$$

It was shown by McFadden (1974) that,

$$\Pr(Y_i = k) = \frac{e^{\boldsymbol{\beta}^\top \mathbf{x}_{ik}}}{\sum_{l=1}^K e^{\boldsymbol{\beta}^\top \mathbf{x}_{il}}}.$$

This model is known as the **conditional logit model** (also referred to as multinomial logit often). The function above is also often called **softmax** function.

Special Case of the model (Multinomial Logistic Regression):

Assuming only individual specific data and suppose we want to categorize the individual into one of several categories it gives rise to a multinomial logistic regression model.

$$\Pr(Y_i = k | \mathbf{x}_i) = \frac{e^{\boldsymbol{\beta}_k^\top \mathbf{x}_i}}{\sum_{l=1}^K e^{\boldsymbol{\beta}_l^\top \mathbf{x}_i}}$$

Here $\boldsymbol{\beta}_k$ is a vector of weights corresponding to category k .

1.3 Assessing the quality of fit

1. Akaike's Information Criterion:

$$AIC = -2LL(\hat{\beta}) + 2p$$

where LL : Log-likelihood, p : no. of parameters. Smaller the AIC, the better.

2. Likelihood ratio index (McFadden's index):

$$\rho := 1 - \frac{LL(\hat{\beta})}{LL(\mathbf{0})}$$

where $\hat{\beta}$ is the estimated value of the parameters. $LL(\mathbf{0})$ refers to the log-likelihood when all the parameters are set to 0 (no model).

This compares the quality and fit of the model in comparison to a model in which all the parameters are equal to 0. The likelihood ratio index ranges from 0 (estimated model is no better than zero parameters) to 1 (estimated model perfectly predicts the choice observed).

3. Percent correctly predicted

This identifies the alternative with the highest probability for each individual observation and determining whether or not this was what the actual choice was.

2 Maximum Likelihood Estimation

Given the observations:

$\mathbf{x}_{ik}$ (attributes of individual i and alternative k).

$$z_{ik} = \begin{cases} 1 & \text{if individual } i \text{ chooses } k \\ 0 & \text{otherwise. (Note } y_i = k \iff z_{ik} = 1) \end{cases}$$

The problem of estimating the β (weights) that maximises the likelihood of the observations is given as:

$$\max_{\beta} \underbrace{L(\beta)}_{\text{Likelihood given } \beta} = \max_{\beta} \prod_{i=1}^n \prod_{k=1}^K \Pr(Y_i = k)^{z_{ik}}$$

Taking logarithms, this problem can be reformulated as maximizing the log-likelihood.

$$\max_{\beta} \underbrace{LL(\beta)} = \max_{\beta} \sum_{i=1}^n \sum_{k=1}^K z_{ik} \log(\Pr(Y_i = k)) = \max_{\beta} \sum_{i=1}^n \sum_{k=1}^K z_{ik} \log \left(\frac{e^{\beta^\top \mathbf{x}_{ik}}}{\sum_{l=1}^K e^{\beta^\top \mathbf{x}_{il}}} \right)$$

This problem can be efficiently solved as the objective function is concave. This is one of the few formulas for choice probabilities where the objective function is known to be concave.

3 Additional material (advanced)

3.1 Derivation of multinomial logit probability

Denote $v_{ik} := \beta^\top x_{ik}, \forall k = 1, \dots, K$ and $i = 1, \dots, n$.

$$\begin{aligned}
\Pr(\text{individual } i \text{ chooses } k^{th} \text{ alternative}) &= \Pr(U_{ik} > U_{il}, \forall l \neq k) \\
&= \Pr(v_{ik} + \epsilon_{ik} \geq v_{il} + \epsilon_{il} \forall l \neq k) \\
&= \Pr(\epsilon_{il} \leq v_{ik} - v_{il} + \epsilon_{ik} \forall l \neq k) \\
&= \int_{-\infty}^{\infty} \Pr(\epsilon_{il} \leq v_{ik} - v_{il} + \epsilon_{ik} | \epsilon_{ik} = y) \underbrace{e^{-y} e^{-e^{-y}}}_{\text{Density of } \epsilon_{ik}} dy \\
&= \int_{-\infty}^{\infty} \underbrace{\prod_{l \neq k} \Pr(\epsilon_{il} \leq v_{ik} - v_{il} + y)}_{\text{From independence of } \epsilon_{il}} e^{-y} e^{-e^{-y}} dy \\
&= \int_{-\infty}^{\infty} \prod_{l \neq k} e^{-e^{-(v_{ik} - v_{il} + y)}} e^{-y} e^{-e^{-y}} dy \\
&= \int_{-\infty}^{\infty} e^{-y} \prod_{l=1}^K e^{-e^{-(v_{ik} - v_{il} + y)}} dy \\
&= \int_{-\infty}^{\infty} e^{-y} e^{-\left(\sum_{l=1}^K e^{-(v_{ik} - v_{il} + y)}\right)} dy \\
&= \int_{-\infty}^{\infty} e^{-y} e^{-e^{-y} \left(\sum_{l=1}^K e^{-(v_{ik} - v_{il})}\right)} dy \\
&= \int_{-\infty}^0 -e^{-t} \sum_{l=1}^K e^{-(v_{ik} - v_{il})} dt \quad (\text{define } t = e^{-y}, dt = -e^{-y} dy) \\
&= \frac{e^{v_{ik}}}{\sum_{l=1}^K e^{v_{il}}}.
\end{aligned}$$

3.1.1 A property of Multinomial Logit Model: Independence of irrelevant alternatives

Observe that for the MNL model we have

$$\frac{\Pr(Y_i = k)}{\Pr(Y_i = l)} = \left(\frac{e^{\beta^\top \mathbf{x}_{ik}}}{\sum_{t=1}^K e^{\beta^\top \mathbf{x}_{it}}} \right) \frac{1}{\frac{e^{\beta^\top \mathbf{x}_{il}}}{\sum_{t=1}^K e^{\beta^\top \mathbf{x}_{it}}}} = e^{\beta^\top (\mathbf{x}_{ik} - \mathbf{x}_{il})},$$
$$\log \frac{\Pr(Y_i = k)}{\Pr(Y_i = l)} = \beta^\top (\mathbf{x}_{ik} - \mathbf{x}_{il})$$

This property of the MNL model is known as the Independence of Irrelevant Alternatives wherein a choice set consisting of two alternatives k and l , adding in a third alternative does not change the ratio of $\Pr(Y_i = k)/\Pr(Y_i = l)$. Namely the new alternate gains share proportionately from the choice share of existing alternatives in the set.

Example: Blue bus/ Red bus.

Alternatives: Car, Red Bus

$\Pr(C) = \Pr(R) = 0.5$ (Suppose a commuter chooses between a car and a red bus with equal probability)

Assume a new blue bus is added and reasonably commuters don't care about the color of the bus.

Alternatives: Car, Red Bus, Blue Bus

$\Pr(C) = 0.5, \Pr(R) = \Pr(B) = 0.25$: Reasonable prediction

$\Pr(C) = \Pr(R) = \Pr(B) = 0.33$: MNL prediction

This happens because blue bus and red bus are perfect substitutes here and not captured by MNL model.

Note this might be a smaller issue in the case of Oscars predictions since it is not clear if nominees are likely to be close substitutes except if an individual receives multiple nominations in a category in the same year or nominated movies might be from similar genres making them closer substitutes.